Experiment 4: simple steps

Friday 23 October, 13:00 to 17:00

Do the boxes in order. Tick each line when it is done. Write every number on the sheet on the last page.

Four rules for the whole day

1. **One sensor at a time.** Gyroscope only for everything that rolls. Accelerometer only for the two flat recordings in Step 3.
2. **Export after every recording.** Click the small icon at the bottom left of the chart, then *Export to CSV*. The file lands in `Documents/iOLab-WorkFiles/export`. Drag it into the folder **4th_experiment/data** on the Mac. Do not rename it. (Other laptop: attach the files in the chat.)
3. **1 second still** at the start of every recording, before anything moves.
4. **Nobody touches the ramp** after Step 3. If it moves, do Step 3 again.

Step 1. Start the iOLab — 13:00, 15 min

- ☐ Take the small USB dongle out of the iOLab (press its metal end). Plug it into the laptop. Switch the iOLab on with the button on its side.
- ☐ Open the iOLab app. Menu **Tools**, then **Acc/Magn/Gyro calibration**. Do what the screen says: six still positions on a flat table.
- ☐ Tick **Gyroscope** only. Write the Hz number from the chart title on the sheet.
- ☐ Press Record. Spin the iOLab fast in your hand around its long side. Press Stop. Look at the top of the biggest curve.

Round peaks above 9 rad/s: all good. A peak that is cut off flat: write the number where it is cut and tell me at once, before you build the ramp.

- ☐ One person draws the sketch on paper now: the ramp from the side (like the picture on page 2) and the iOLab with its two rolls from the front. Photo of it at the end.

Step 2. Weigh and measure the parts (rolls OFF) — 13:15, 15 min

- ☐ Weigh the iOLab alone, dongle out.
- ☐ Weigh roll 1. Weigh roll 2.
- ☐ The hole in each roll: measure the inside diameter twice, turning the ruler or caliper a quarter turn between the two.
- ☐ The outside of each roll: wrap a paper strip once around it, mark where the strip overlaps itself, lay it flat and measure the length. That is the circumference. **Do not** measure across the roll with a ruler.
- ☐ The iOLab: width (short side, about 7.5 cm) and thickness (about 3 cm).
- ☐ The cardboard ring inside one roll: its thickness (about 3 mm).

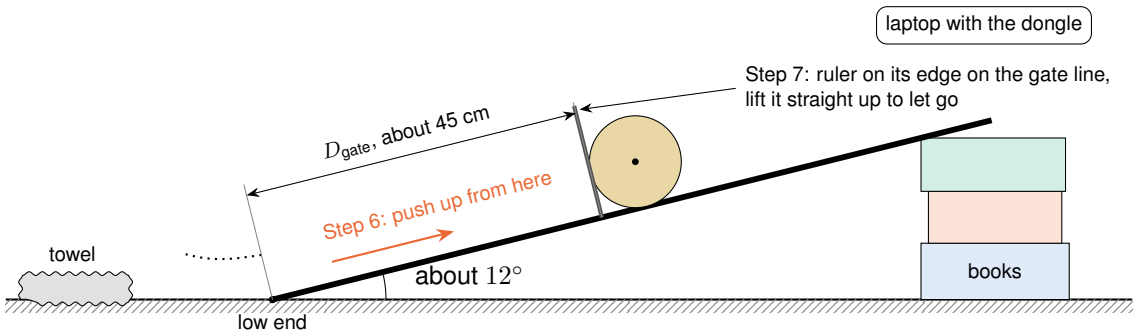
Step 3. Build the ramp and measure the angle — 13:30, 20 min

- ☐ A stiff board, at least 80 cm long. Low end on the table, high end on books. Rise about 21 cm per metre of board (about 12 degrees). Mark a straight line down the middle with tape.
- ☐ Towel on the table, 30 cm past the low end.
- ☐ **Recording A**, **Accelerometer** only, bare iOLab (no rolls). Press Record, then hold each position still for 4 seconds:
  1. iOLab standing up against a wall or a box, the long side straight up (y up; the axes are printed on the iOLab).
  2. Upside down (y down).
  3. Flat on the board near the bottom, long side along the board, y pointing *up* the slope. Hands off.
  4. Turned around, same face down, y pointing *down* the slope.
  5. Do 3 and 4 again in the middle of the board, then again near the top.

  Press Stop. Export.

- ☐ **Recording B**, Accelerometer only: iOLab flat on the table next to the ramp, y pointing toward the high end, 4 seconds. Turn it around, 4 seconds. Stop. Export.
- ☐ Ruler check: two tape marks on the board, 80 cm apart. Measure the height of the top of the board above the table at both marks, ruler held straight up.
- ☐ Gate mark: a tape line across the board 45 cm up from the low end. Measure along the board from this line to the low end edge: D_{gate}, to the millimetre.

From now on, nobody touches the ramp.



The ramp from the side (angle drawn larger than real). The circle is the iOLab with its rolls, resting against the gate.

Step 4. Put the rolls on — 13:50, 5 min

- ☐ Pick two rolls whose circumferences are within 2 mm of each other.
- ☐ Push a roll onto each end of the iOLab up to the small notch. **Not past the notch.** Same depth at both ends.
- ☐ Press the loose end of the tape flat on both rolls.
- ☐ Weigh the whole thing: M.
- ☐ Measure its length, outer edge of roll 1 to outer edge of roll 2.
- ☐ Gyroscope only. Record and roll it gently along the table. The y curve is big, x and z stay near zero, and it rolls straight. Write on the sheet if y goes plus or minus when it rolls away from you.

Step 5. Calibration roll, 3 times — 13:55, 15 min

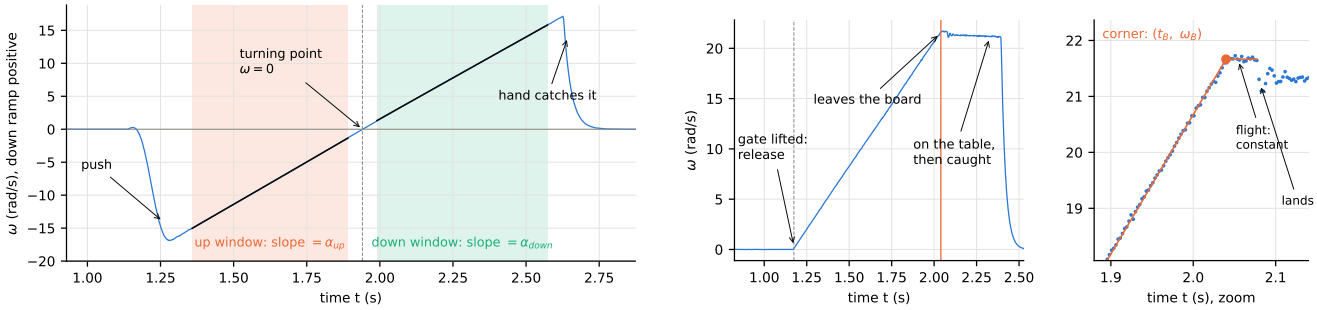
- ☐ Draw a pen line across the outside of roll 1. Stick tape on the table and draw a start line on it.
- ☐ Gyroscope only. Set the body down with the pen line on the start line. Record. 1 second still.
- ☐ Push it slowly with one finger on the middle of the iOLab, in a straight line, for exactly **3 full turns**: stop when the pen line touches the table for the third time. Mark that spot. 1 second still. Stop.
- ☐ Measure from the start line to the mark (about 1 m) to the millimetre. Write it. Export.
- ☐ Do it three times.

Step 6. Method N: push up, let it roll back, 6 to 8 times — 14:10, 25 min

- ☐ Gyroscope only. Body at rest on the tape line, about 10 cm above the low end, rolls square to the board. Record. 1 second still.
- ☐ One smooth push up the board with one finger, then take the hand away at once. It should go 30 to 50 cm up, stop and roll back.
- ☐ Catch it before it reaches the low end. Stop. Export.
- ☐ Look at the graph. Good: after the push, one straight sloping line that passes through zero (left picture below). Bad: it drifted sideways more than 2 cm, hit something, or the line is bumpy. Do a bad run again and write “bad” on the sheet.

Step 7. Method E: let go from the gate, 6 to 8 times — 14:35, 20 min

- ☐ Hold a ruler on its edge across the board on the gate line. Rest the body against it from above.
- ☐ Gyroscope only. Record. 1 second still.
- ☐ Lift the ruler straight up, fast. Do not push the body.
- ☐ It rolls down, flies off the end and lands on the table. Stop. Export.
- ☐ Good: a straight rise from zero, then a sharp corner where it leaves the board (right picture below).



Left: a good Step 6 run. Right: a good Step 7 run. Your numbers will differ a little.

Step 8. Send me everything — 14:55

- ☐ All CSV files in **4th_experiment/data** (or attached in the chat).
- ☐ The sheet on the last page, typed in the chat or as a photo.
- ☐ Which runs were bad.
- ☐ Keep the ramp standing until I say the data is fine. That takes about 10 minutes.

Step 9. The report — 15:15 to 16:55

- ☐ I write the report from your data and send it to you.
- ☐ You put in the photo of the hand sketch and read every line.
- ☐ Page 4: I send a draft of the three reflections. Put them in your own words.
- ☐ Save as PDF and upload on DTU Learn by 16:55. Do not wait until 17:00.

The sheet

fill it in as you go, then send it to me

Step 1 Gyroscope rate _____ Hz Peaks round? yes / no, cut at _____ rad/s

Step 2 iOLab _____ g roll 1 _____ g roll 2 _____ g scale step _____ g

Hole diameter roll 1: _____ mm and _____ mm roll 2: _____ mm and _____ mm

Circumference roll 1 _____ mm roll 2 _____ mm

iOLab width _____ mm thickness _____ mm cardboard ring _____ mm

Hole measured with: caliper / ruler

Step 3 Ruler marks apart _____ cm height low _____ cm height high _____ cm

D_{gate} _____ cm

Step 4 Whole body M _____ g length _____ mm y rolling away from you: plus / minus

Step 5 Distance for 3 turns: roll 1 _____ cm roll 2 _____ cm roll 3 _____ cm

Step 6 Runs: 1 2 3 4 5 6 7 8 9 10 (circle the bad ones)

Step 7 Runs: 1 2 3 4 5 6 7 8 9 10 (circle the bad ones)

Notes

If something goes wrong

- Curve cut off flat in Step 6 or 7: the gyroscope is at its limit. Make the ramp lower (about 6 degrees) and redo Step 3. Tell me.
- The body rolls in a curve: the two rolls differ. Swap one roll, or push it to the same depth as the other.
- A bump once every turn in the curve: the tape end is loose, or the iOLab sits crooked in a roll. Press the tape flat, turn the roll a little on the iOLab.
- It slips instead of rolling: the ramp is too steep or dusty. Wipe the board.
- Running late: do 5 runs in Step 6 and 5 in Step 7. Never skip Step 3 or Step 5.